

ELECTRONICS LABORATORY REPORT

Automatic Temperature Control and Cooling System

Project Name:

Automatic Temperature Control and Cooling System

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1. Description

1.1 Project Overview

This project presents the design and simulation of an Automatic Temperature Control and Cooling System developed using Proteus Design Suite. The primary objective of the circuit is to monitor the ambient temperature and autonomously activate a cooling fan — represented in the simulation by a DC motor — once the measured temperature surpasses a pre-set threshold value. The system operates on a closed-loop "Sense, Compare, and Act" automation principle, eliminating the need for any human intervention during normal operation.

1.2 Component List and Purpose

The following components were selected for this design, each fulfilling a specific and critical role within the circuit:

Reference	Component	Value / Model	Function
RT1	NTC Thermistor	10k @ 25°C	Temperature sensing
RV1	Potentiometer	10k	Threshold adjustment
R1	Resistor	33k	Voltage divider (fixed arm)
R2	Resistor	1k	Base current limiter
R3	Resistor	10k	LED current limiter
C1	Electrolytic Capacitor	100 µF	Power line filtering
U1:A	Op-Amp	LM358N	Voltage comparator
Q1	NPN Transistor	BC547	Driver / electronic switch
D1	LED	Red (LED-RED)	Visual status indicator
D2	Rectifier Diode	1N4007	Flyback protection
RL1	Relay	12V	Load-side isolation switch
M1	DC Motor	12V	Cooling fan representation

1.3 System Logic

The circuit follows two clearly defined operating states based on the measured temperature:

State 1 — Temperature Below Threshold (Cooling OFF):

When the ambient temperature is low or at room temperature, the NTC thermistor maintains a high resistance. This results in a high voltage at the inverting input (Pin 2) of the LM358N Op-Amp. Since $V_- > V_+$, the comparator output is driven LOW (0V). Consequently, the BC547 transistor remains off, the relay stays open, the motor is stationary, and the LED is unlit.

State 2 — Temperature Above Threshold (Cooling ON):

As ambient temperature increases, the NTC thermistor resistance drops, causing the voltage at Pin 2 to fall. Once $V_+ > V_-$ (Pin 3 reference voltage exceeds Pin 2), the Op-Amp output switches HIGH. This signal passes through R2 to the base of Q1, turning the transistor on. Current flows through the relay coil, closing the relay contacts and starting the motor. Simultaneously, D1 lights up as a visual alert.

1.4 Protection and Stability Features

- Flyback Diode (D2 — 1N4007): Placed in parallel with the relay coil to suppress reverse-voltage spikes generated when the relay de-energises. This protects the BC547 transistor from breakdown.
- Decoupling Capacitor (C1 — 100 μ F): Connected across the 12V supply rail to filter out voltage dips and electrical noise produced by the motor and relay switching, ensuring the Op-Amp reference remains stable.

2. Schema

The complete circuit schematic was designed in Proteus Design Suite. It is presented below, showing all components, interconnections, supply rails, and ground references. The design is divided into functional stages that flow logically from left to right: Sensing Stage → Reference Stage → Decision Stage → Driver Stage → Output/Load Stage.

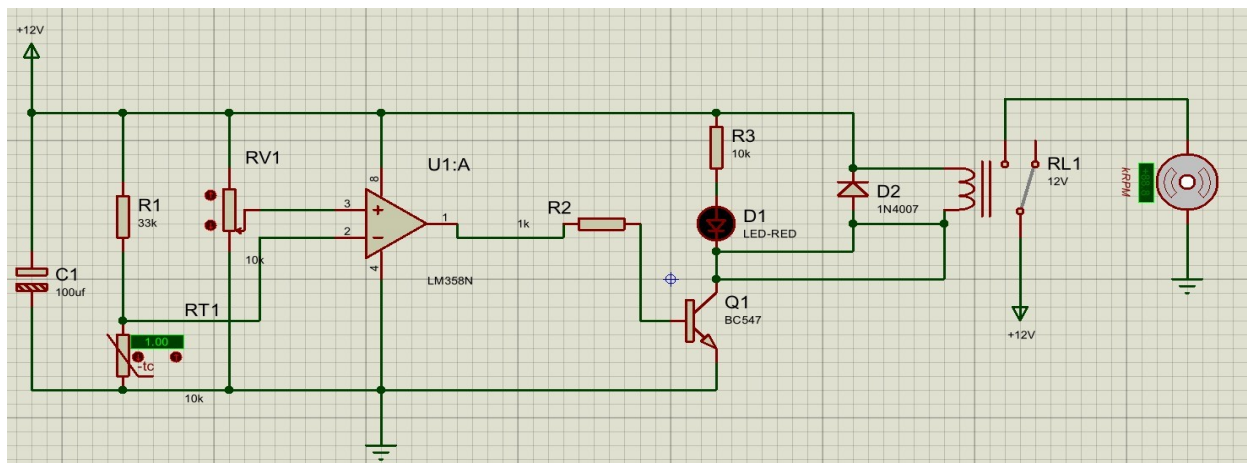


Figure 1: Complete Circuit Schematic — Automatic Temperature Control and Cooling System (Proteus)

2.1 Stage-by-Stage Description

Reading the schematic from right to left, the functional stages are as follows:

- Output Stage: DC Motor (M1) connected to the Normally Open (NO) contacts of relay RL1, powered from the +12V rail.
- Switching Stage: RL1 12V Relay isolates the high-current motor side from the low-voltage control circuitry.
- Protection & Indicator Stage: 1N4007 flyback diode (D2) across the relay coil; Red LED (D1) in series with R3 for visual feedback.
- Driver Stage: BC547 NPN transistor (Q1) with R2 (1k) as base current-limiting resistor.
- Decision Stage: LM358N Op-Amp (U1:A) acting as a voltage comparator; Pin 8 (+12V) and Pin 4 (GND) for supply.
- Reference Stage: RV1 potentiometer sets the reference (threshold) voltage at Pin 3 (non-inverting input).
- Sensing Stage: Voltage divider formed by R1 (33k) and RT1 NTC thermistor delivers a temperature-proportional voltage to Pin 2 (inverting input).
- Power Filtering: C1 (100 μ F) at the supply input stabilises the entire circuit against switching transients.

3. Calculations

3.1 Voltage Divider — Sensing Network

The NTC thermistor (RT1) and fixed resistor R1 (33k Ω) form a voltage divider. The voltage at the inverting input of the Op-Amp is given by:

$$V_- = V_{t^e} \times R_{_NTC} / (R1 + R_{_NTC})$$

At room temperature (25°C), $R_{_NTC} = 10\text{k}\Omega$:

$$V_- = 12 \times 10,000 / (33,000 + 10,000) = 12 \times 0.2326 \approx 2.79 \text{ V}$$

As temperature rises and $R_{_NTC}$ decreases (e.g. to 3k Ω at ~60°C):

$$V_- = 12 \times 3,000 / (33,000 + 3,000) = 12 \times 0.0833 \approx 1.00 \text{ V}$$

3.2 Reference Voltage — Potentiometer (RV1)

The RV1 potentiometer (10k Ω) sets the threshold (reference) voltage at Pin 3 of the Op-Amp. Its output voltage is adjustable from 0V to +12V depending on wiper position:

$$V_+ = V_{t^e} \times (R_{_wiper} / R_{_total}) = 12 \times (0 \text{ to } 10,000) / 10,000$$

For a typical set point of ~1.5V:

$$R_{_wiper} = 1.5 / 12 \times 10,000 = 1,250 \text{ } \Omega \text{ (12.5\% of full rotation)}$$

3.3 Base Current — BC547 Transistor (Q1)

The base current required to saturate Q1 is limited by R2 (1k Ω). Assuming the Op-Amp output swings to approximately 10V (high state) and $V_{_BE} = 0.7\text{V}$:

$$I_{_B} = (V_{_out} - V_{_BE}) / R2 = (10 - 0.7) / 1,000 = 9.3 \text{ mA}$$

With a minimum h_{FE} of 110 for BC547, the maximum collector current that can be driven into saturation:

$$I_C(\max) = h_{FE} \times I_B = 110 \times 9.3 \text{ mA} \approx 1,023 \text{ mA} = 1.02 \text{ A}$$

The relay coil typically draws 50–80 mA, well within the transistor's safe operating range.

3.4 LED Current — Indicator Stage

R3 (10k Ω) limits the current through the Red LED (D1). With a forward voltage drop $V_f \approx 2.0\text{V}$:

$$I_{LED} = (V_{CC} - V_f) / R3 = (12 - 2.0) / 10,000 = 1.0 \text{ mA}$$

This current is sufficient to illuminate the LED while keeping power dissipation minimal (10 mW).

4. Simulation

4.1 Simulation Environment

The circuit was fully simulated using Proteus Design Suite 8 (Professional Edition). Proteus was selected for this project due to its accurate SPICE-based simulation engine and its comprehensive library of real-world components including the LM358N Op-Amp, BC547 transistor, NTC thermistor, and relay models.

4.2 Simulation Procedure

The following steps were performed during the simulation phase:

- All components were placed on the Proteus schematic editor and connected according to the circuit design.
- The NTC thermistor (RT1) was set to a nominal resistance of 10k Ω at 25°C with a negative temperature coefficient of -1.00 (as shown in the component properties).
- The potentiometer RV1 was adjusted to set a reference voltage of approximately 1.5V at Pin 3 of the LM358N.
- The simulation was run at room temperature first to confirm the motor remained stationary and the LED was off.
- The thermistor temperature was then increased interactively using the Proteus interactive component controls until the thermistor resistance fell below the switching threshold.

4.3 Simulation Results

The simulation confirmed the expected circuit behaviour under both operating conditions:

Condition	Op-Amp Output (Pin 1)	System State
$T < \text{Threshold}$ (RT1 high)	~0V (LOW)	Motor OFF, LED OFF
$T > \text{Threshold}$ (RT1 low)	~10V (HIGH)	Motor ON, LED ON

Condition	Op-Amp Output (Pin 1)	System State
Relay switching	Flyback spike suppressed	D2 conducting briefly

The flyback diode (D2) successfully suppressed the reverse voltage spike generated by the relay coil during de-energisation. The 100µF capacitor (C1) maintained a steady supply voltage throughout the switching events, preventing the Op-Amp comparator from glitching.

4.4 Observed Waveforms

During simulation, the following voltage levels were observed at key nodes:

- Pin 2 (V₋): Approximately 2.8V at room temperature, falling below 1.5V as temperature rose above the threshold.
- Pin 3 (V₊): Fixed reference of ~1.5V set by RV1.
- Pin 1 (Op-Amp Output): Clearly switched between 0V and ~10V as the temperature crossed the threshold.
- Q1 Collector: Dropped from +12V to near 0V (saturation) when the transistor turned ON.

5. Troubleshooting

During the design and simulation phases, several potential issues were identified and resolved. The following table summarises the most common faults and their solutions:

Symptom	Probable Cause	Solution
Motor never turns on	RV1 threshold set too high; V+ always below V ₋	Reduce RV1 wiper setting to lower reference voltage
Motor always on regardless of temperature	RV1 set too low; V+ always above V ₋	Increase RV1 setting to raise reference voltage
Transistor Q1 overheating	Insufficient base current; Q1 not in full saturation	Verify R2 value; check Op-Amp output swing voltage
Op-Amp output unstable / flickering	No hysteresis; noisy thermistor signal at threshold	Add positive feedback resistor for Schmitt trigger hysteresis
Relay chattering repeatedly	Same as above; oscillation at the switching threshold	Introduce hysteresis or a small RC filter at Pin 2
LED not lighting up	R3 too large; or D1 connected in reverse polarity	Check LED orientation; reduce R3 if necessary
Transistor destroyed after relay switch-off	Flyback diode D2 missing or reversed	Confirm D2 cathode is towards +12V rail

Symptom	Probable Cause	Solution
Circuit resets when motor starts	Power supply cannot handle inrush current; no C1	Ensure C1 (100μF) is present across supply rail

5.1 Design Recommendations

- Hysteresis: For a more stable switching action in real-world environments with noisy signals, a small positive feedback resistor (typically 1MΩ) between Pin 1 and Pin 3 of the LM358N would introduce hysteresis, preventing chattering near the threshold voltage.
- Temperature Calibration: The potentiometer RV1 must be calibrated using a known temperature reference (e.g. a thermometer placed near RT1) to set the activation threshold accurately.
- PCB Layout: In a physical implementation, keep the thermistor away from heat-generating components such as the relay and transistor to ensure accurate ambient temperature sensing.

6. Questions

The following theoretical and analytical questions are provided to test understanding of the circuit design, component behaviour, and underlying electronic principles.

Question 1

What is the fundamental operating principle of an NTC (Negative Temperature Coefficient) thermistor, and how does it differ from a PTC (Positive Temperature Coefficient) thermistor? Explain how the NTC characteristic is exploited in this circuit design.

Question 2

The LM358N Op-Amp is used as an open-loop voltage comparator in this circuit. Describe the conditions under which its output transitions from LOW to HIGH. Write the mathematical inequality that governs this switching decision, referencing the non-inverting (V₊) and inverting (V₋) input voltages.

Question 3

Calculate the voltage at the inverting input (Pin 2) of the Op-Amp when the NTC thermistor resistance is 5kΩ. Given: Supply voltage V_{CC} = 12V, R1 = 33kΩ, R_{NTC} = 5kΩ. Using the formula: $V_- = V_{CC} \times R_{NTC} / (R1 + R_{NTC})$.

Question 4

Explain the role of the flyback diode (D2 — 1N4007) in this circuit. What physical phenomenon does it protect against, and what would happen to the BC547 transistor if this diode were absent from the circuit?

Question 5

Why is the BC547 NPN transistor necessary in this circuit? Could the LM358N Op-Amp directly drive the relay coil without the transistor? Support your answer with reference to typical Op-Amp output current specifications and the relay coil current requirements.

Question 6

The decoupling capacitor C1 (100 μ F) is placed at the power supply input. Explain why power supply decoupling is particularly important in circuits that include relay and motor loads. What type of interference does C1 mitigate?

Question 7

Propose two real-world applications of this temperature control circuit, other than the computer CPU cooling example discussed in the project. For each application, briefly describe how the circuit behaviour would need to be adapted, if at all, for that specific use case.

Question 8

In the current design, the motor either runs at full speed or is completely off (binary control). Describe how the circuit could be modified — either by adding components or replacing the Op-Amp comparator with a different control strategy — to achieve variable motor speed proportional to the measured temperature (analogue/PWM control).

— End of Report —